

2.3 INDUSTRIAL APPLICATIONS

As mentioned in Section 1.3.1, serial robots are mainly used in industries for welding, assembling, machining, etc. As more and more robots are designed for a specific task, they can be classified based on their tasks, e.g., ‘welding robot’, ‘assembly robot’, etc. Since these robots are designed for a specified work, they cannot be made readily adaptable to other applications. For example, an assembly robot cannot be used for welding purposes unless a major change is done. Note that some suppliers provide a complete robot system, e.g., for welding, with welding equipment and other material-handling facilities like turntables, etc., as an integrated unit. Such integrated robotic system is also called a ‘welding robot’ even though its discrete manipulator unit could be adapted to a variety of tasks. Some robots are specifically designed for heavy load manipulation, and are labeled as ‘heavy-duty robots’.

There are certain industrial tasks which robots do very well like material transfer, machine loading, spot welding, continuous arc welding, spray coating, material removing, cutting, assembling, inspection, sorting, cleaning and polishing parts, and a dozen more specialized tasks. In this section, some of these applications will be examined as per their actions of end-effector (Groover et al., 2012), namely, the following:

Material Handling In these applications, the robot grasps an object, e.g., a machined component, with the help of a gripper, and moves it to another location, say, on a conveyor belt.

Processing Here, the robot uses a tool, e.g., an electrode, rather than a gripper in order to perform some processing task, say, welding.

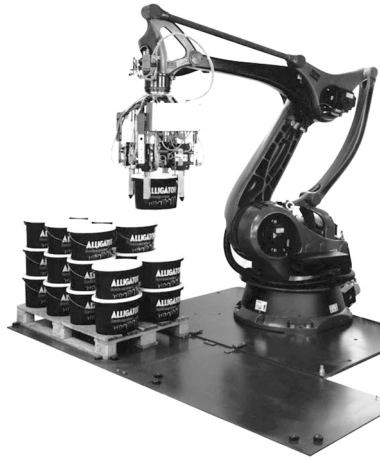
Assembling These tasks are more complex than other two, as the robot has to interact with a fixture or another robot in order to put together two or more components or sub-assemblies to complete an assembly task.

2.3.1 Material Handling

During the sixties and the seventies, automation affected primarily the manufacturing process but not the auxiliary functions such as handling, set-up, loading, unloading, etc. The time spent to transfer a workpiece from one station to the next is still high. Up to 95 percent of the time involved in manufacturing a part is composed of transfer and waiting time, and only about 5 percent of the total time is actual processing. Whereas the processing time has been reduced considerably by automation, much less progress has been made in handling and loading. The fully automatic systems that were developed for mass production (e.g., transfer lines in the automobile industry) are rigid (hard automation) and not suitable for batch production (in the order of 50 to 100,000 parts annually). A more flexible automation technology that takes into account frequent changes in production is needed for this category of industrial production, which accounts for about 75 percent of the manufactured parts. A new solution was offered to the handling and machine-tool loading of small and medium-size parts with the development of industrial robots. Actually, loading and unloading of machine tools are the major applications of robots. Robots are utilized to load and unload machine tools in two basic configurations:

- (a) A robot tending a single machine, and
- (b) a robot serving several machines.

The first configuration is applied when typical machining times per part is short. The second configuration is applicable when a chain of operations must be executed to complete a part. For many applications, the cylindrical, polar, and revolute types are equally suitable whereas in some light-duty work in restricted areas, rectangular types may be used. For material-handling applications, point-to-point control is necessary. A typical material-handling robot is shown in Fig. 2.17.

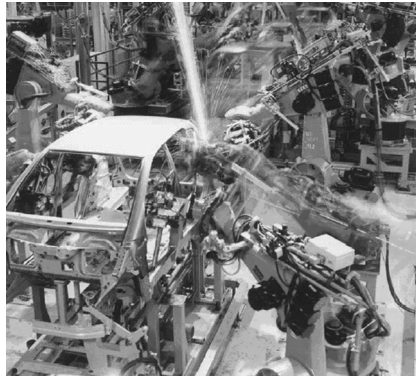


[Courtesy: <http://www.packworld.com/machinery/palletizing/hassia-usa-inc-robotic-palletizer/>]

Fig. 2.17 A pick-and-place material-handling robot

2.3.2 Welding (Processing)

Welding is a manufacturing process in which two metal pieces are joined usually by heating and fusing. The welding operations performed by robots are thermal processes in which the metal pieces are joined by melting or fusing their contacting surfaces. These processes can be grouped under two classes, namely, where no filler material is added to the joint interface, and in which a filler material of the same parent metal is added. Accordingly, there are two types of welding operations performed by the robots, namely, spot and arc welding, respectively, as shown in Fig. 2.18.

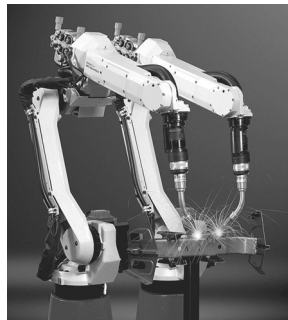


[Courtesy: http://www.kukarobotics.com/en/solutions/solutions_search/L_R142_Body_shop_Mercedes-Benz_A_Class.htm]

(a) Spot welding of car parts in an assembly plant

1. Spot Welding

In spot welding, two metal pieces are joined only at certain points by melting or fusing their contacting surfaces. The required heat is generated by the passage of an electric current through the metals at the point where they are to be joined. It is frequently used in the automotive industries to join thin sheet metals. A spot-welding robot has to carry the



[Courtesy: <http://www.cnc-arena.com/en/newmachines/907/jointed-arm-robot/fanucrobotics/m-10ia>]

(b) A pair of robots performing arc welding

Fig. 2.18 Welding robots

welding gun, which consists of electrodes, cables required to conduct the high current, and sometimes a water-cooling system for the electrodes. Since the welding gun is relatively heavy (10 to 80 kg), most of the spot-welding robots are hydraulically powered as dc-motor-driven robots cannot handle such heavy loads. The control system for spot-welding robots is of a point-to-point (PTP) type with reasonable positional accuracy and repeatability of ± 1 mm. This repeatability of a robot is much higher than that of a human welder. Hence, a robotic welding is preferred over manual one for uniformity of the products at hand. Further, the operation of robotized spot-welding is very fast.

Accuracy vs. Repeatability

Accuracy is defined as the difference between the commanded and actual positions attained by the robot end-effector, whereas repeatability is defined as the closeness between the actual positions attained by the end-effector for the same command.

Spot-welding robots, as shown in Fig. 2.18 (a) for a car assembly, affected the car industry most. However, they can be used in fabrication of metal products, domestic appliances, metal furniture, containers which do not require liquid-tight joints, etc. The first spot-welding robots were installed in 1969 at a General Motor's plant for welding of car bodies (Engelberger, 1980). Since then spot-welding robots have proved to be very profitable. Moreover, parts of the robots can be suspended from the ceiling, which saves expensive floor space. Several robots can operate simultaneously on the same car body, which increases the efficiency of the assembly line. A human operator might miss a weld or make it in an incorrect location, and, therefore, many times extra welds are added at the design stage. With robot operation, the work is consistent and all the welds are placed in the right location and, therefore, the required body strength can be achieved by specifying fewer welds. A typical assembly line produces between 50 to 90 cars per hour, and the work is performed while the car bodies are continuously moving on conveyors, which means that the weld locations specified by the task programs should be synchronized with the velocity of the assembly line.

2. Arc Welding Arc welding falls in the category in which two metals pieces are joined along a continuous path by adding a filler material of the same type as the parent metal. The required heat is provided by an electric arc generated between the electrode and the metals. It is needed, for example, in sealing a container against leakage. While most robotic-arc welding uses a consumable wire electrode, e.g., in MIG welding, with an automatic wire feeder, welding with non-consumable tungsten electrodes with shielding gas, as in TIG welding, is also in use.

In arc welding, the robot shown in Fig. 2.18(b) uses the welding gun as a tool. The consumable electrode, which provides the filler material, is in the form of a wire (coiled on a drum) of the same composition as the material to be welded. Wire diameters of 1/32 to 3/16 in (0.8 to 4.8 mm) are commonly used. The wire is automatically fed by a motor with adjustable speed at a preset rate that is determined by the arc voltage. The wire feed increases with an increase in the voltage applied between the work and the electrode. This voltage can be monitored and used to maintain a constant arc length by varying the speed of the motor which feeds the wire. In order to keep the electrode cooler and permit higher currents to be used, the

shielding gas flows in a tube along the electrode. The tube is terminated in a nozzle at the end of the gun from which the gas flows into the arc region. Robotic welding systems sometimes use water-cooled guns. The weight of the welding gun is usually not heavy (unless the water-cooled type is used) and, therefore, dc servomotor-driven robots are typically used in arc welding, although hydraulically drive robot are also sometimes found. Welding speeds range from about 10 to over 120 in/min (0.25 to 3 m/min). The welding current usually ranges between 100 and 300 A, but with the larger electrodes (3/16 in or about 4.75 mm) the current may be as high as 1200 A, resulting in a very deep penetration of the weld. The control system for robots in arc welding is usually of a Continuous Path (CP) type. Nevertheless, PTP control systems are also used.

2.3.3 Spray Painting (Processing)

Spray painting is also a kind of processing used in manufacturing industries. The unhealthy and unpleasant environment of the painting booth in industry made this process an ideal candidate for the application of robots. The solvent materials that are used in spray painting are toxic, and, therefore, the operators must be protected by masks and be provided with fresh-air ventilation. The painting area must be dust-free and temperature-controlled, and consequently the painting booth is small in size and inconvenient for the operators. Furthermore, the noise arising from the air discharge through the painting nozzles can cause irreversible damage to the ears. For all these reasons, spray painting became one of the first applications of robots. The requirement for robots in spray painting are different from those of other robot applications, and, therefore, many robot manufacturers offer a robot dedicated to this one application. The spray-painting robots are of CP capability and have the following characteristics:

- (a) high level of manipulator dexterity,
- (b) large working volume for small-base manipulator,
- (c) compact wrist,
- (d) small payload, and
- (e) low accuracy and repeatability.

Figure 2.19 shows the spray painting of a car chassis by robots. The painting robot must be able to carry any type of spray gun. Spray guns, however, are light weight and, therefore, painting robots are designed for small payloads (e.g., 1 kg). Finally, the requirements for repeatability and resolution are the least severe in painting robots. Therefore, a repeatability of 2 mm throughout the working volume is regarded as sufficient for spray-painting robots.

2.3.4 Machining (Processing)

Machining is another type of manufacturing process. There are five basic types of machine tools to perform machining, namely, drilling machine, lathe or turning machine, milling machine, shaper, and grinder. Out of all these machining operations, only drilling is being successfully done with robots, and mainly in the aircraft industry. Another application related to machining which is performed by robots is deburring of metal parts.



Fig. 2.19 Spray-painting robots with car chassis

1. Drilling Robots can replace the manual operators if the template hole is provided with a chamfered guide. The gripper of the robot holds a portable drill and guides from hole to hole. At each hole, a fixed drill cycle is performed, and then the robot moves the drill to the next hole. Programming the robot to perform the task is quite simple. Since drilling is a PTP operation, the manual teaching method is appropriate. Figure 2.20(a) shows a robotized drilling operation.



(a) Drilling



(b) Deburring

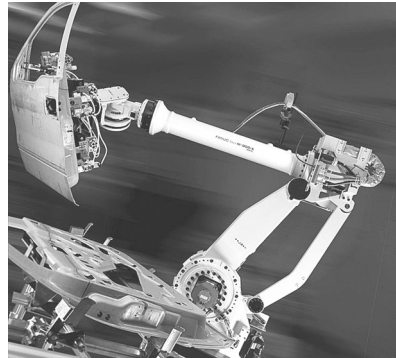
Fig. 2.20 Machining robots

2. Deburring Burrs are generated almost always when machining is performed on metal parts. Burrs can be generated between a machined surface and a raw surface or at the intersection between two machined surfaces. The removal of these burrs is an expensive operation. Most deburring is performed manually by workers equipped with appropriate tools. By closely following the manual method, the industrial robots can solve most deburring problems. Figure 2.20(b) shows deburring operation by a robot. There are two basic ways to perform robotized deburring. If the part is relatively lightweight, it can be picked up by the robot and brought to the deburring tool. If the part is heavy then the robot holds the tool. The support of the tool is very important, whether it is held by the robot or mounted on the work table. In both cases, the relative motion between the tool and the part is of a CP type with high repeatability (approximately 0.2 mm) and highly controlled speed. Therefore, deburring is one of the most difficult tasks for robots.

2.3.5 Assembling

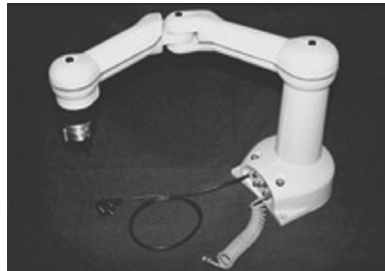
Assembling with industrial robots is mainly used for small products such as electrical switches and small motors. Robotized assembly systems are programmable and, therefore, provide a cost-effective solution for the assembly of small batch sizes and for batches containing different products. Although industrial robots require the same fixtures, feeders, and other equipment for positioning the parts as conventional assembly machines, simpler workpiece feeder and fixtures may be used because of the robots' programmability feature. Furthermore, tactile or optical sensors may be added to the assembly robot to tackle more complex assembly tasks. Some assembly tasks require the participation of more than one robot. In order to reduce the cost per arm, there are systems in which several Cartesian arms can use the same base and share the same controller.

Assembly robots, e.g., the one similar to Fig. 2.21, can be designed in any coordinate system: Cartesian, cylindrical, spherical, or articulated. However, many tasks require only vertical assembly motions, such as the assembly of printed circuit boards. For these applications, the 4-DOF robot shown in Fig. 2.22 can be adequate. Its arm has two articulated motions, and the wrist has two axes of motion: a linear vertical displacement and a roll movement. This robot can pick up parts located on the horizontal plane, bring them to the assembly location, orient them with the roll motion of the wrist, and finally insert them in a vertical motion. This class of robot is known as the SCARA type which was first developed in Japan.



[Courtesy: <http://www.fanucrobotics.com>]

Fig. 2.21 Assembly robot



[Courtesy: <http://www.systemantics.com/products.htm>]

Fig. 2.22 A 4-DOF robot from Systemantics India

Example 2.3 Robot Considerations for an Application

There are several points which should be taken into account while choosing a robot for an application. Some of the important ones are listed below:

- Degrees-of-freedom (DOF) of the robot for the planned application
- Robot work volume
- Payload capacity
- Type of end-effector, i.e., gripper for material handling, etc.
- For grippers, its total DOF, number of fingers or jaws
- Actuator type, i.e., electric, hydraulic, or pneumatic
- Point-to-point or PTP (mostly used in material-handling applications) or continuous-path or CP (welding, spraying and similar applications)

- Programming language
 - Safety features
 - Initial investment and running cost
 - Payback period
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SUMMARY

In this chapter, several subsystems of a robotic system are presented. A robot is classified based on its workspace, controlling method, etc. Industrial applications of the robots in the area of manufacturing, etc., are also presented in detail. Classification of a robot, either based on workspace or application, will enable a user to select/order a robot for his or her requirements.

EXERCISES

- 2.1 What are the different subsystems of a robotic system?
- 2.2 Why are transmissions used in a robot manipulator?
- 2.3 When should a belt-pulley be used over gears?
- 2.4 Show an arrangement of conveying boxes.
- 2.5 What are the advantages of using a mechanism over other transmission arrangements?
- 2.6 What are different ways of classifying a robot?
- 2.7 Why are ADC and DAC required in robots?
- 2.8 What kind of robot is suitable for painting a shaft? Why?
- 2.9 What is the shape of the workspace of the SCARA robot similar to Fig. 2.22? Draw it.
- 2.10 Why are some workspaces empty?
- 2.11 What is the full form of PUMA?
- 2.12 What type of basic arm does the PUMA robot have?
- 2.13 What type of control is suitable in welding operations?
- 2.14 Name typical industrial applications of robots.
- 2.15 What is the repeatability demanded in a spraying robot?

WEB-BASED EXERCISES

Based on Web search, find the answers to the following questions:

- 2.16 Name some robot manufacturers and their robot programming languages.
- 2.17 What are the typical gear ratios used in industrial robots?
- 2.18 Find some industrial applications where gantry robots are used.
- 2.19 Name some industries which employ SCARA robots and why.
- 2.20 Find out the model numbers of KUKA, ABB, and Kawasaki robots for the payload capacity of up to 15 kg.
- 2.21 What are the accuracy and repeatability of the above robots?
- 2.22 What are typical prices of industrial robots used in welding of car bodies?